

Hsin-Yuan (Samuel) Lee

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Research Interests: Mechanistic interpretability, LLM evaluation, neurocomputational vision, LLM scientific reasoning

Education

University of California, Los Angeles, Los Angeles, CA

Sep 2026 – Jun 2028 (Expected)

Master of Science (M.S.) in Computer Science

University of California, San Diego, La Jolla, CA

Sep 2022 – Jun 2026

B.S. in Data Science; B.S. in Cognitive Science (Machine Learning & Neural Computation)

GPA: 3.92/4.00

Cum Laude; Provost's Honors (8 quarters).

Publications

When AI Explains AI: Understanding and Mitigating the Interpretability Lottery

H.-Y. Lee, S. Chang, T. Wang, Z. Wang. *ICLR 2027*, under review.

HypoEvolve: Genetic Algorithms Enable Multi-Agent LLMs to Discover Scientific Hypotheses

J. Liu, M. Hu, J. Chen, J. Kong, P. Jagannatha, Y. Gao, D. Pratt, H.-Y. Lee, Z. Hu, T. Ideker, W. Wang, E. P. Xing, Z. Wang. *ICLR 2027*, under review. arXiv:2609.15938.

A Mechanistic Explanation for the Inverted Face Effect

A. Tahan*, H.-Y. Lee*, K. Fleischer*, N. Kachappilly*, X. Chen*, G. W. Cottrell. *CogSci 2026*. *Co-first authors.

Research Experience

Reliability of Automated Interpretability (Dr. Zhen Wang)

Mar 2026 – Present

Student Researcher; First Author, ICLR 2027 Submission

- Built an LLM auto-interpretability pipeline (Claude Haiku explainer, GPT-4o-mini scorer) that labels Gemma-2-2B sparse autoencoder (SAE) features, and reran it **10,000 times**, including 15 runs on each of 150 features.
- No feature kept a single label across reruns; scores barely changed and were unrelated to label instability ($r = -0.01$).
- Ran a $2 \times 2 \times 2$ factorial design that traced most instability to which examples the explainer is shown.
- Developed a zero-cost mitigation: fixing each feature's examples to its top- k contexts cut label instability **47%** (95% CI 43–51%), holding on held-out features.

Gary's Unbelievable Research Unit (Prof. Garrison Cottrell), UC San Diego

Aug 2025 – Jun 2026

Undergraduate Research Assistant

- Trained a ResNet-18 with fixation-based sampling, log-polar transforms, and salience-weighted preprocessing on **76,800 images** of 384 face, dog, and object identities.
- Reproduced the human Face Inversion Effect, matching human accuracy within **6%** in all four Yin (1969) conditions.
- Built reproducible evaluation scripts to measure inversion robustness under varying levels of expertise.
- Co-first author of the resulting paper, published at CogSci 2026.

HypoEvolve, HDSI Senior Capstone (Prof. Zhiting Hu, Dr. Zhen Wang), UC San Diego

Sep 2025 – Mar 2026

Capstone Researcher

- Contributed to a multi-agent LLM system that evolves scientific hypotheses with a genetic-algorithm loop.
- Owned the Type-2 Diabetes validation track: best GA configuration scored **0.291** mean top-1 Open Targets vs. 0.164 single-shot LLM and 0.122 random ($n=10$); on a matched base model, GA lifted it from 0.008 to 0.192.

Qualcomm Institute (Calit2) (Dr. Neil Smith), UC San Diego

Aug 2025 – Jun 2026

Research and Development Intern

- Wrote the core of a real-time, LLM-driven 3D avatar system in **C++ and Unreal Engine 5** Blueprints, integrating text-to-speech and gesture generation; demoed to **20+ visitors**.
- Deployed a modular LLM-TTS-animation pipeline on FastAPI and Kubernetes, cutting avatar latency **25%**.
- Ran gesture-synchrony and prosody-alignment experiments supporting early-stage embodied AI research.

Industry Experience

Realtek Semiconductor Corp., Hsinchu, Taiwan

Jul 2026 – Sep 2026

Software Engineering Intern

- Built two Jenkins-integrated Bash/Python automation tools (~9,900 lines) replacing a manual Git/Gerrit repo-mirroring workflow, cutting full-repository sync from **~3 hours to ~25 min** across **400+ repositories** used by 30 engineers.
- Built a SHA-256 checksum diff engine across git worktrees for binary/LFS repositories that `git diff` can't compare.
- Automated cross-branch cherry-picks via `git patch-id` matching, with one-click Gerrit pushes and CSV audit trails.
- Wrote SKILL.md specs that let a Claude Code agent drive the workflow; packaged the tools as a versioned skill hub.

Advantech Co., Ltd., Taipei, Taiwan

Jun 2023 – Sep 2023

Software Development Intern

- Built a retrieval-augmented generation (RAG) assistant on Azure OpenAI and Azure AI Search, deployed as a Discourse forum bot and web chatbot, cutting response time from days to under 20 seconds.
- Designed a punctuation-aware Mandarin tokenizer, lifting ambiguous-query retrieval accuracy from **34% to 76%**.
- Automated document-ingestion and update pipelines that kept the search index continuously refreshed.
- Worked with the frontend team to ship the user interface; won the **Best Technical Intern Project** award.

Selected Projects

Deep Climate Learning (Kaggle competition)

Mar 2025 – Jun 2025

- Built CNN-MLP (precipitation) and ClimaX transformer (temperature) models in PyTorch Lightning.
- Designed a custom area-weighted loss that reduced regional error, especially at high-variance latitudes; ranked **2nd of 83 teams**, cutting RMSE by **46%** vs. baseline.

Weak-to-Strong Generalization on MATH (with Dr. Zhen Wang)

Oct 2025 – Dec 2025

- Tested whether weak models' few-shot demos can supervise stronger models on MATH via in-context learning.
- Benchmarked 5 OpenAI models (gpt-5-mini, o3-mini, gpt-4.1-mini, gpt-4.1-nano, gpt-4o-mini) in an async pipeline.
- Found Performance Gap Recovered (PGR) usually near 1, but more weak examples did not always help and noisy ones hurt stability.

Block Blast Reinforcement Learning Agent

Jan 2026 – Mar 2026

- Formulated Block Blast as an MDP with action masking and a shaped reward for line clears, combos, and game over.
- Built an AlphaZero-style agent: a dual-head CNN (board and piece encoders, 192-action policy head, tanh value head) guiding PUCT Monte Carlo Tree Search, trained by self-play over 1,000 episodes.
- Averaged **264.0 points vs. 73.8** for a random agent over 500 evaluation games (~3.9×), surviving 31.3 vs. 16.4 moves.

Clickbait Detection with Transformers (team of 3)

Oct 2025 – Dec 2025

- Fine-tuned BERT, DistilBERT, and RoBERTa-base on 32,000 news headlines against TF-IDF classical baselines.
- Tuned hyperparameters with a two-stage Weights & Biases grid search: 2,000-sample sweeps, then full-data retraining.
- RoBERTa reached **0.876 test F1** (90.8% accuracy) vs. 0.859 (89.0%) for the best baseline, logistic regression.

Liver Tumor Segmentation (team of 3)

Jan 2023 – Jun 2023

- Trained a U-Net on CT scans (68% accuracy) and built a GUI for uploading scans.

Teaching and Leadership

ACM AI @ UC San Diego

May 2024 – Jun 2026

Director of Competitions (May 2025 – Jun 2026); Competition Developer (May 2024 – May 2025)

- Led two campus-wide AI competitions (StarChess.ai, Blockography.ai) with **150+ participants**.
- Built the competition infrastructure: game environments, evaluation scripts, leaderboards, and Discord support.
- Introduced beginner-friendly tracks and starter notebooks, increasing novice participation by **40%**.

Cognitive Science Department, UC San Diego

Jan 2024 – Jun 2024

Instructional Assistant: COGS 9 Intro to Data Science; COGS 14B Intro to Statistical Analysis

- Led weekly sections and office hours on coding and statistics; graded assignments and projects with written feedback.

Technical Skills

Languages: Python, C++, JavaScript, Bash

Research Areas: Mechanistic Interpretability, Sparse Autoencoders, LLM Evaluation, AI Agents, Computer Vision, NLP

Methods: Experimental Design, Statistical Analysis, Deep Learning, Transformers, CNNs, RL (MCTS)

Tools: PyTorch, PyTorch Lightning, HuggingFace, Weights & Biases, scikit-learn, NumPy, pandas, Docker, Git, Gerrit